Simple Linear Regression

MGMT 670

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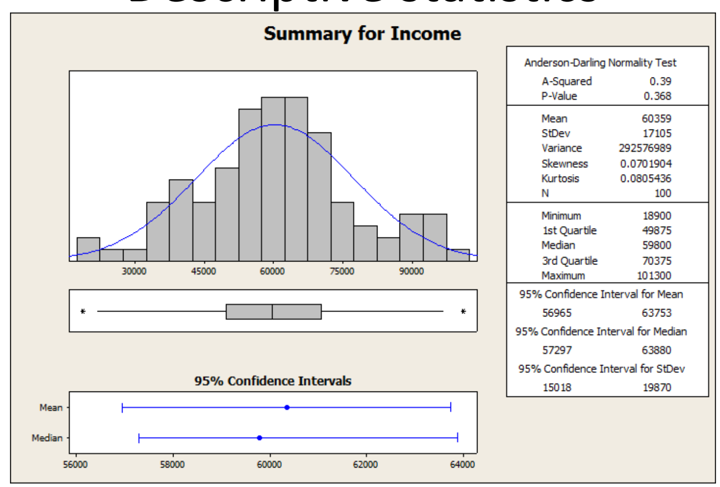
Topics

- Estimating parameters
  - Regression equation
  - Explained; unexplained variations(SST, SSR, SSE)
  - R square
- Tests: T-test/F-test
- Prediction: PI vs CI
- Residual Analysis
- Common Interview Questions:
  - https://www.geeksforgeeks.org/machine-learning/top-linear-regression-interview-questions/
  - https://github.com/Devinterview-io/linear-regression-interview-questions

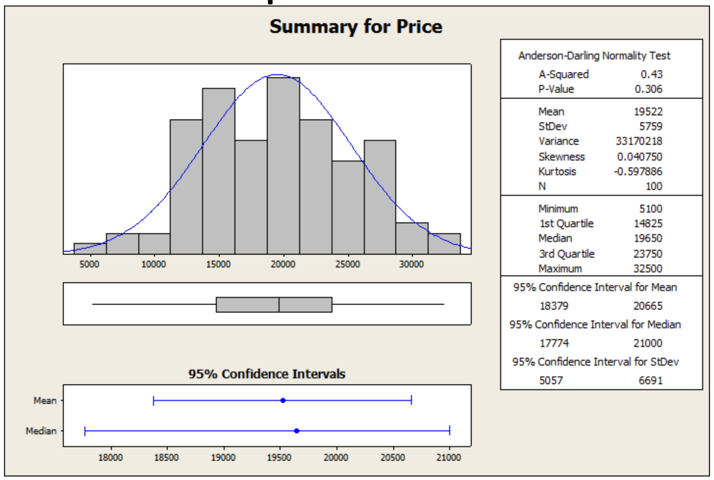
Example: Car dealership

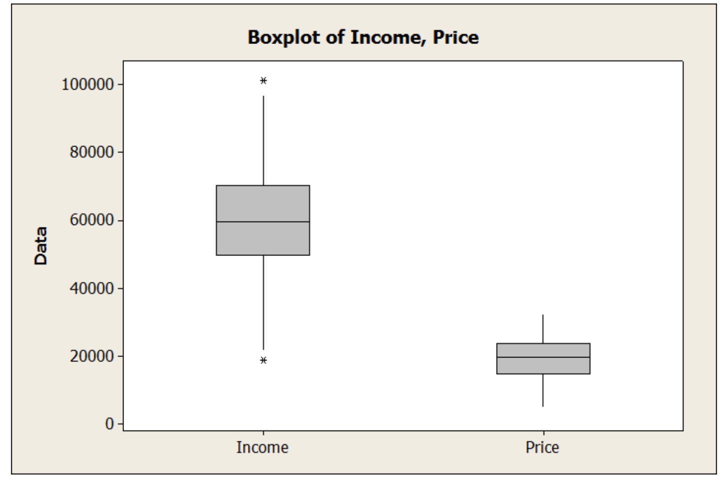
- Car dealership is opening at a new location, would like to estimate the number and type of cars to be stocked.
- **Rule of thumb** based on financial advices and previous data
  - People buy cars around 1/3 of annual income
  - An extra \$3K increase in annual income spend around \$1K more on car
- We obtain recent data (*cardealership.csv*) from other location, and want to analyze this data

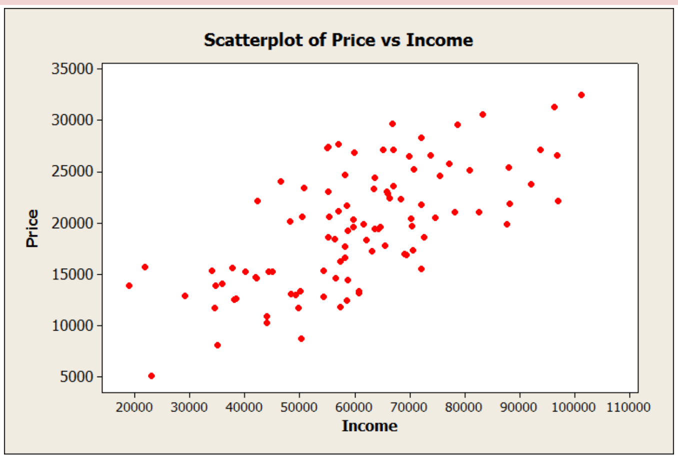
Descriptive statistics



Descriptive statistics







correlation of Income and Price = 0.676

- How can we check if rule of thumb is 1/3?
- What else can we do with this data?

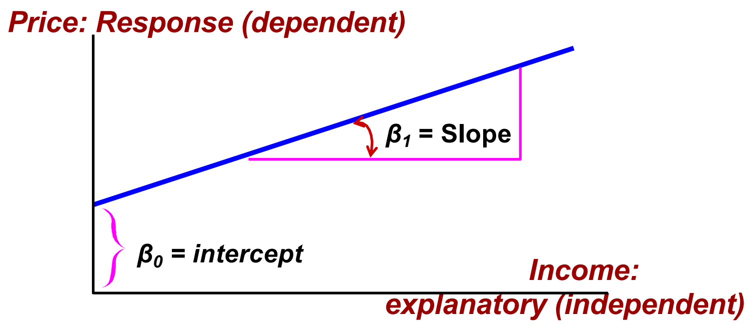
Regression Models

- Describe a relationship between a
  - **response (dependent)** variable and
  - at least one **explanatory (independent)** variable.
- Used for prediction
  - e.g., sales vs. promotion activities
- **NOT** used for
  - e.g., Euro to US Dollar exchange rate

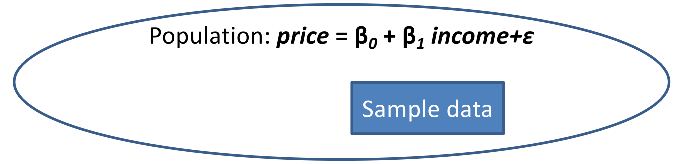
Simple Regression Model

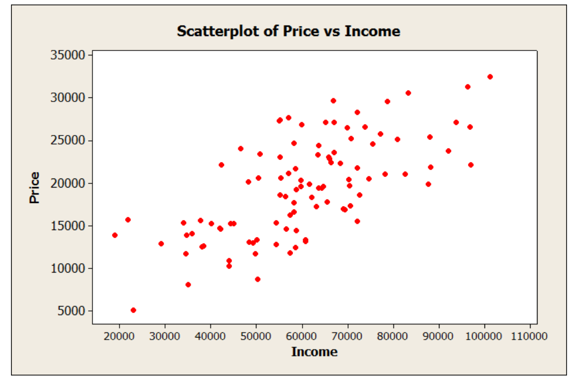
$$Model: price = \beta_0 + \beta_1 \times income + \varepsilon, \quad \varepsilon \sim \text{Normal}(0, \sigma)$$

standard deviation σ



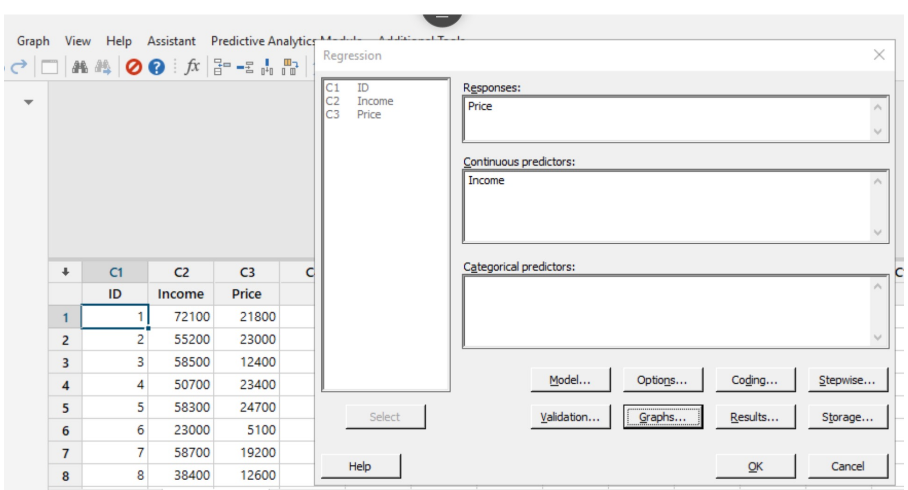
What does the model mean in this example?





We want to use the data to estimate β_0, β_1, and σ !

Minitab: Stat->Regression->Fit Regression Model



Using Minitab

Regression Analysis: Price versus Income

Regression Equation

Price = 5788 + 0.2275 Income

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	5788	1572	3.68	0.000	
Income	0.2275	0.0251	9.08	0.000	1.00

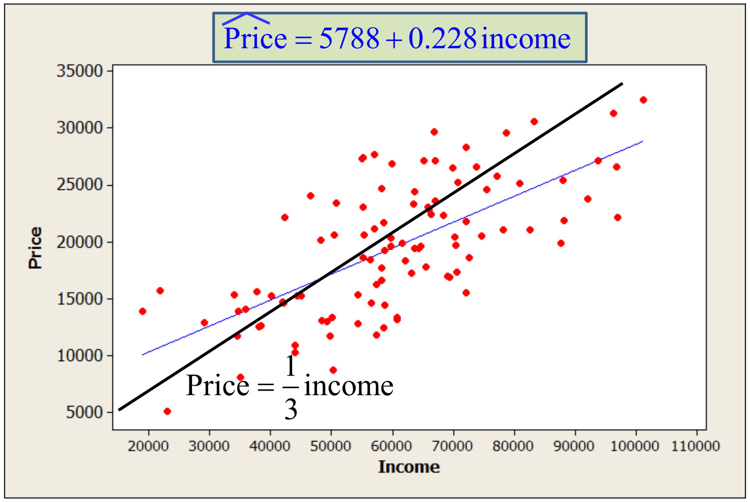
Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
4266.86	45.67%	45.11%	43.61%

Analysis of Variance

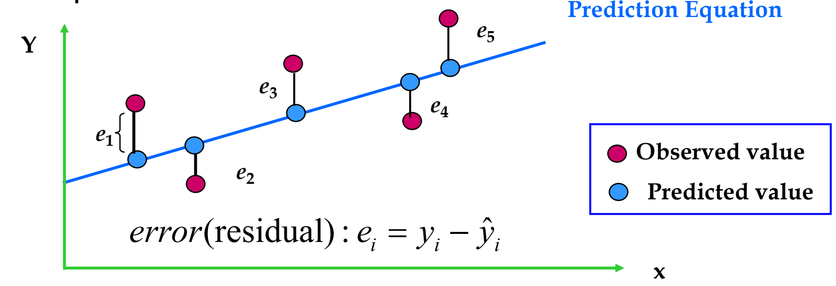
Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	1499656198	1499656198	82.37	0.000
Income	1	1499656198	1499656198	82.37	0.000
Error	98	1784195402	18206076		
Lack-of-Fit	91	1593145402	17507092	0.64	0.842
Pure Error	7	191050000	27292857		
Total	99	3283851600			

Why is the regression line better than the rule of thumb?



How is the best-fit line calculated

Least Squares Method



minimize the sum of square errors $\sum_{1}^{n}(y_i - \hat{y}_i)^2$

Best-fit line formula

- Sum of squares and cross products

$$SS_X = \sum (x_i - \bar{x})^2$$
$$SS_Y = \sum (y_i - \bar{y})^2$$
$$SS_{XY} = \sum (x_i - \bar{x})(y_i - \bar{y})$$

- Regression estimators

$$b_1 = \frac{SS_{XY}}{SS_X} \quad \text{estimator for } \beta_1$$
$$b_0 = \bar{y} - b_1\bar{x} \quad \text{estimator for } \beta_0$$

- Regression equation

$$\hat{Y} = b_0 + b_1 X$$

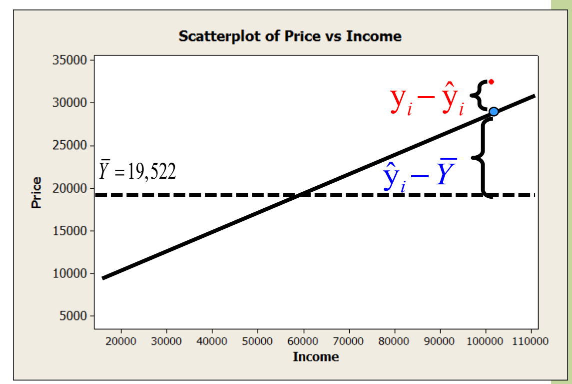
Explained, Unexplained Variations

unexplained variation

$$SSE = \sum_{1}^{n}(y_i - \hat{y}_i)^2$$

explained variation

$$SSR = \sum_{1}^{n}(\hat{y}_i - \bar{Y})^2$$



total variation

$$SST = SSR + SSE$$

$$SST = \sum_{i=1}^{n}(Y_i - \bar{Y})^2$$

$$R^2 = SSR/SST$$

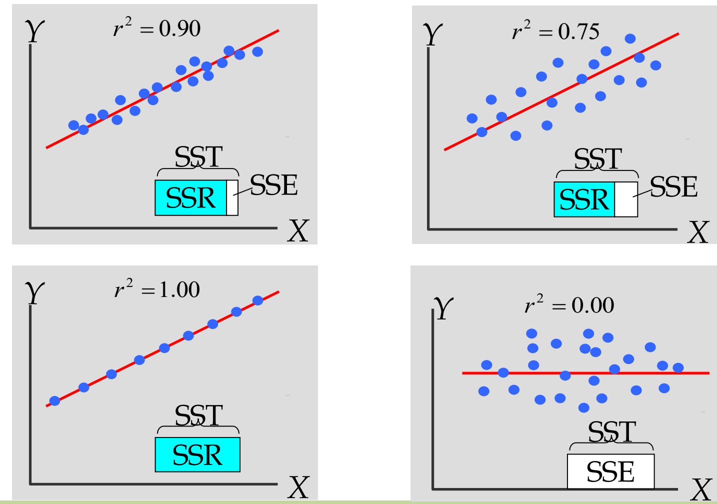
- The most expensive car 32,500 with income 101300. How much does it contribute to SST, SSE and SSR?

- The least expensive car 5100 with income 23000. How much does it contribute to SST, SSE and SSR?

- Using Minitab,
  - SST=
  - SSR=
  - SSE=
  - R-square=

Coefficient of Determination

Note: $r = \sqrt{r^2}$ or $-\sqrt{r^2}$



Testing Significance in Regression

Regression Model: $Y = \beta_0 + \beta_1 X + \varepsilon$

- Testing a linear relationship between X and Y

  (In other words: Does X explain Y?)

  Example: does income levels explain car prices?

- Hypotheses

  H_0: $\beta_1 = 0$ (No linear relationship)

  H_a: $\beta_1 \neq 0$ (Linear relationship)

- Test Procedures
  - *t*-Test

Formula for sampling distribution of b_1

- Variance of population regression error

$$\boldsymbol{Y = \beta_0 + \beta_1 X + \varepsilon, \quad \varepsilon \sim \text{Normal}(0, \sigma^2)}$$

  – Estimator of β_1 is b_1

  – Unbiased Estimator of σ^2

$$s^2 = \frac{\sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2}{n-2} = \frac{SSE}{n-2} = MSE$$

Degree of freedom
n - #of parameters

- Standard Error of $\boldsymbol{b_1}$

$$\boldsymbol{s_{b_1} = \frac{s}{\sqrt{SS_X}}}$$

t-Test

- Test Statistic (5% level of significance)

$$t = \frac{b_1 - 0}{s_{b_1}} = \frac{0.22754 - 0}{0.02507} = 9.08$$

$$t_{(\alpha/2, n-2)} = ?$$

Coefficients

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Conclusion

Reject H_0: $\beta_1 = 0$ and conclude that there is a significant linear relationship between X and Y.

t-Test

What if H_0: $\beta_1 = 1/3$; H_a: $\beta_1 \neq 1/3$?

- Test Statistic (5% level of significance)
- Conclusion:

F-Test in **simple regression**:
(same result as t-test in simple regression)

- **Mean Square Regression**

$$\text{MSR} = \frac{\text{SSR}}{\text{number of independent variables}} = \frac{SSR}{1}$$

- **Mean Square Error**

$$\text{MSE} = \frac{\text{SSE}}{n\text{-number of regression parameters}} = \frac{SSE}{n-2} \quad = s^2$$

- **Test Statistic**

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{SSR}{SSE} \times (\text{n}-2)$$

Compare with critical value $F_{(\alpha, 1, n-2)}$

Conclusion:

If F-test> F-critical value, REJECT NULL hypothesis.

T statistic vs. F statistic

- In a simple regression model, there is only one independent variable. Thus, we have:

H_0: $\beta_1 = 0$

	Test statistics	Two tailed Critical Value at 5%
T-test	9.08	1.984
F-test	82.37=9.08^2	1.948^2=3.936

- In simple regression, tests are identical, and test statistics are related by a square relationship

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P-value?
Conclusion?

Prediction

- Auto dealer: A new customer with income 50K
  - Confidence interval (CI): What's the average price of cars bought **by people with same income?**
  - Prediction interval (PI): What's the actual price of the car that **this person** will buy?
  - PI not only considers the uncertainty in estimating mean (like CI) but also includes the inherent variability of individual data points around mean.

Minitab Procedure for Prediction

Prediction of new observation (income = 50,000)

a. Select Options

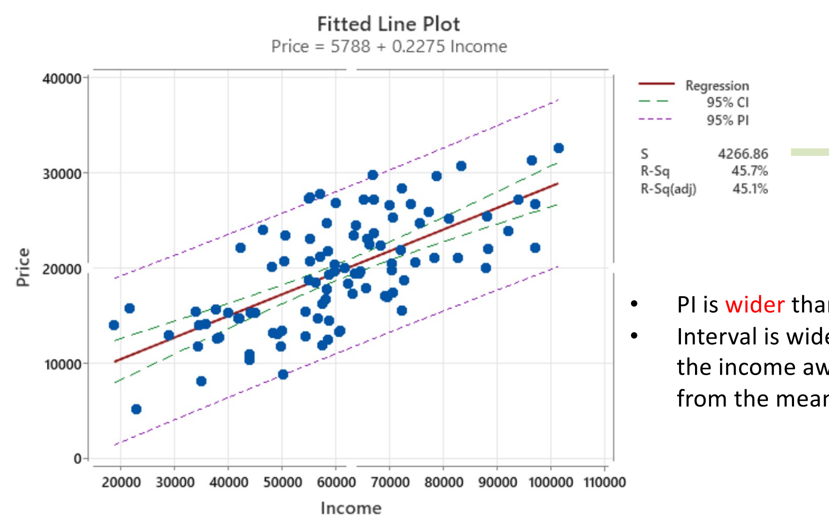
b. Enter the new value and specify the confidence level

Predicted Values for New Observations

New Obs	Fit	SE Fit	95% CI	95% PI
1	17165	500	(16174, 18156)	(8640, 25690)

Values of Predictors for New Observations

New Obs	Income
1	50000



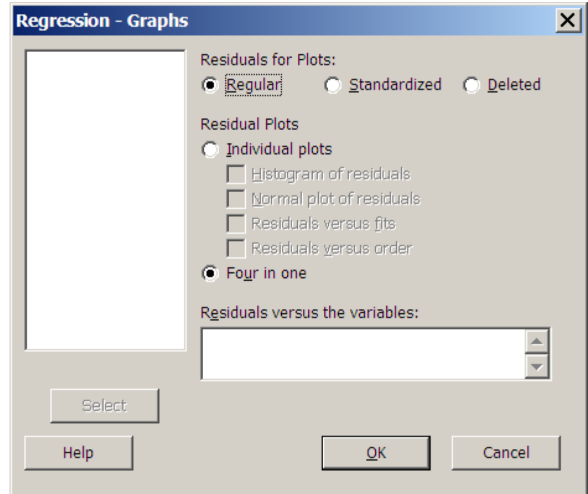
- PI is wider than CI
- Interval is wider if the income away from the mean

- Stat -> Regression -> Fitted line plot...
-Specify response and predictor
-Select Options... select Display confidence interval prediction interval.

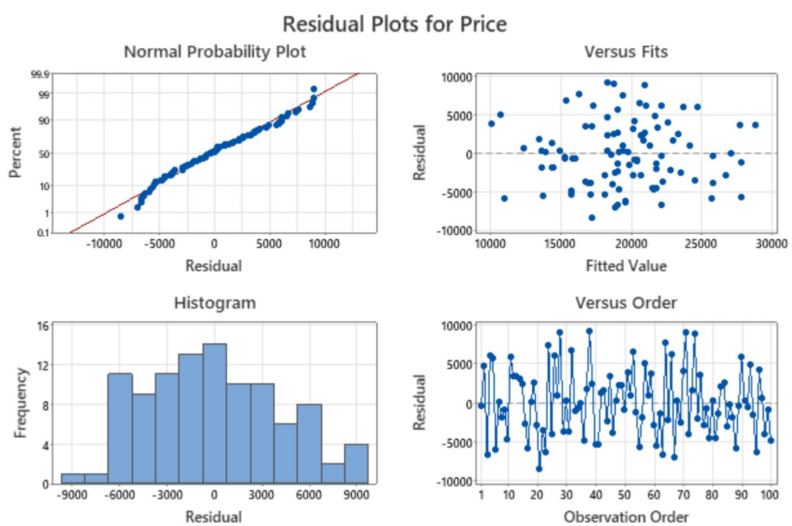
Residual Analysis

- Residual, $e_i = Y_i - \hat{Y}_i$, where $\hat{Y}_i = b_0 + b_1 X_i$
- Examine functional forms
- Evaluate violation of assumptions
  - Normality
  - Homoscedasticity (constant variance)
  - Independence of errors

Creating Residual Plots



Residual Plots: Car-dealership



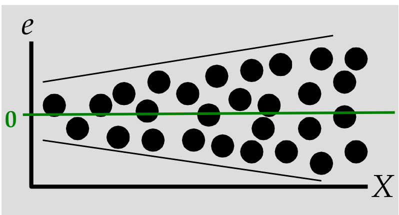
Assessing Normality

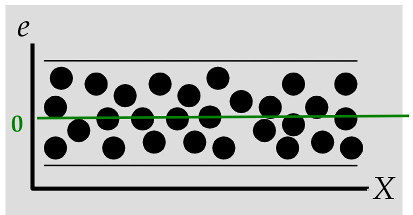
- Examine whether or not the observations follow a normal distribution
- Normal Probability Plot
  - Plot of the observed data values with their corresponding probabilities in normal scale or standardized normal quantile values
  - Assess by line shape (Look for a straight line)

https://en.wikipedia.org/wiki/Normal_probability_plot

Homoscedasticity

Heteroscedasticity | Correct Specification

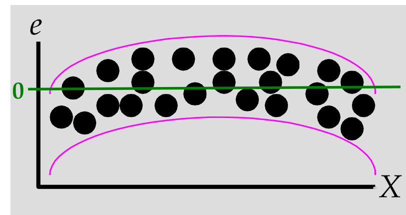


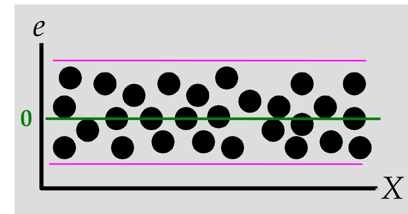


Fan-shaped -> Transform Y

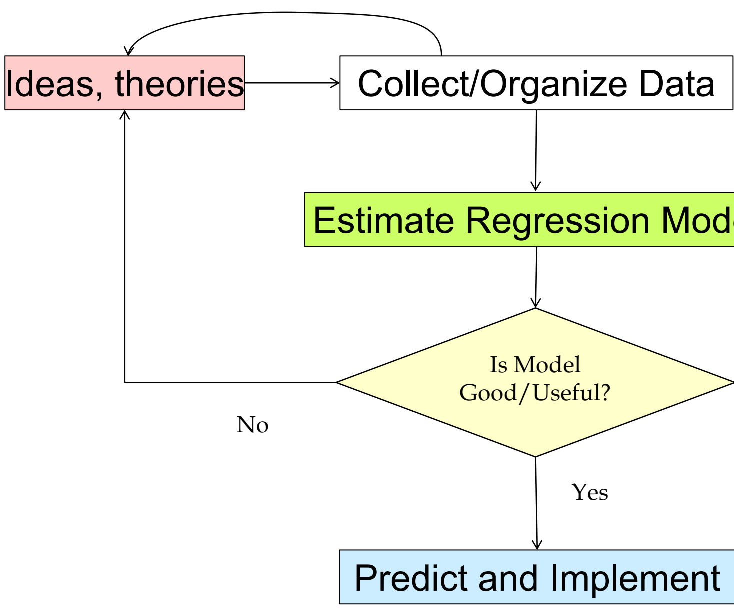
Residual plot for functional forms

Nonlinearity | **Correct Specification**





Transform X
(e.g., Add X^2)



Summary

- Estimating parameters
  - Regression equation
  - Explained; unexplained variations(SST, SSR, SSE)
  - R square
- Tests
  - T-test/F-test
- Prediction: PI vs CI
- Residual Analysis